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UNIT III: Organizing Information & Navigation Systems

3.1 Organizing Information

Definition: Grouping and arranging content into logical categories so users can find and understand it.

- **Goal:** findability and understanding; reduce information overload.
- **Organization system** = organization *schemes* (shared characteristics used to group items) + organization *structures* (relationships between items and groups).
- **Organization vs navigation:** organization = how content is grouped; navigation = how users move through it.
- Supports two user needs: **known-item search** (find a specific thing) and **exploratory browsing** (open-ended discovery).
- Good organization is invisible; users notice it only when they **cannot** find something.
- **Why it matters:** poor organization causes failed searches, wasted time, lost trust and higher support cost.
- **Types of information need:** a known item, a few good items, everything on a topic (exhaustive), re-finding something seen before.
- Organization happens at several levels: whole site, section, page and individual content chunk.

3.2 Organizational Challenges

Definition: Obstacles that make it hard to build one clear organization system for content.

- **Ambiguity:** words and categories have more than one meaning; classification is subjective.
- **Heterogeneity:** mixed content types, formats and levels of granularity (a page vs a whole database).
- **Differences in perspective:** users and departments think about content differently (users do not know the org chart).
- **Internal politics:** departments compete for visibility and control of the structure.
- **Ambiguity example:** one word can mean different things to different users, so a single category name may fail.
- **Heterogeneity:** combining documents, images, applications and databases under one scheme is hard; *granularity* (level of detail) must be decided.
- **Perspective:** organize by what users look for, not by how the company is structured.
- **Ways to cope:** research users and stakeholders, use more than one scheme, get stakeholder agreement early, test the design.

Fig 3.1: Four organizational challenges

3.3 Organizing Websites and Intranets

Definition: Choosing organization *schemes* (how items are grouped) and *structures* (how groups relate).

- **Exact schemes:** clear, mutually exclusive divisions; easy to design and use (Alphabetical, Chronological, Geographical).
- **Ambiguous schemes:** based on subjective categories; harder to design and maintain but better for browsing (Topic, Task, Audience, Metaphor-driven, Hybrid).
- **Organization structures:**
  - **Hierarchy** (top-down tree): simple, familiar; balance breadth vs depth.
  - **Database model**: structured records with fields (metadata); supports search and dynamic views.
  - **Hypertext**: non-linear links between pages; flexible, but can disorient users.

Ambiguous scheme	Organizes by	Example
Topic	Subject matter	Science, Sports, Business
Task	Actions users perform	Pay bill, Apply, Track order
Audience	User group	Students, Faculty, Alumni
Metaphor-driven	Familiar concept	Shopping cart, Desktop, Library
Hybrid	Mix of schemes	Often confusing; avoid inside one list

- **Exact scheme examples:** alphabetical (phone book, glossary), chronological (news archive, event list), geographical (branch locations, maps).
- **Hierarchy design tips:** categories should be mutually exclusive; prefer **broad and shallow** over narrow and deep; group items in small chunks (about 7 plus or minus 2); avoid too many levels.
- **Database model** suits large, dynamic content such as product catalogs; each record has fields (metadata) that can be searched, sorted and filtered.
- **Hypertext** adds links across the hierarchy for associations; it complements, not replaces, a hierarchy.

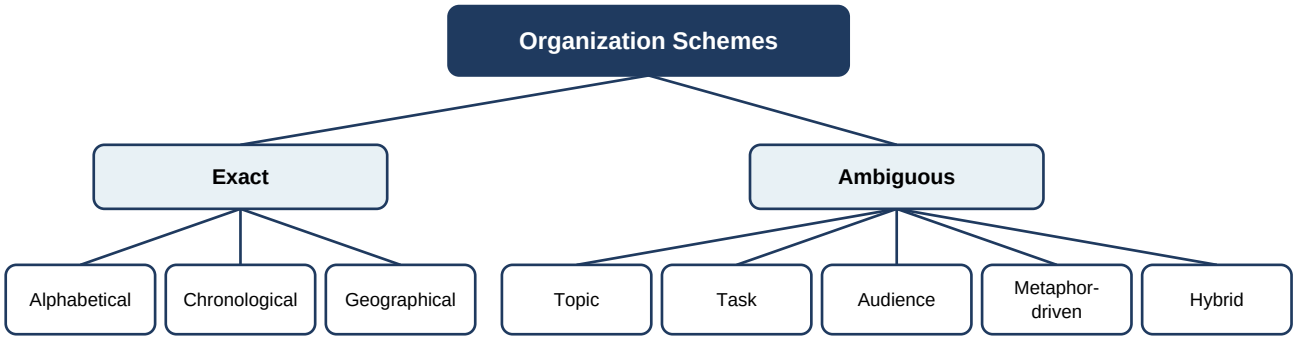
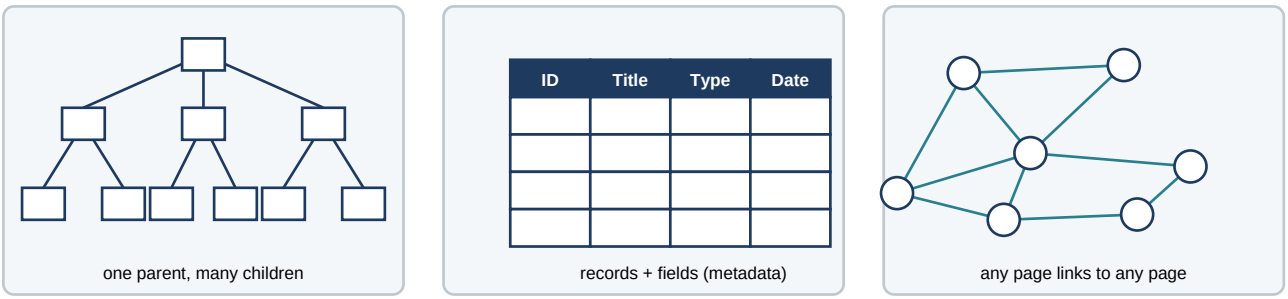


Fig 3.2: Organization schemes



Hierarchy (tree) **Database model** **Hypertext (web)**

Fig 3.3: Organization structures

	Exact schemes	Ambiguous schemes
Design	Easy	Hard (subjective)
Maintenance	Easy	Harder
Best for	Known-item search	Browsing / exploring

3.4 Creating Cohesive Organization Systems

Definition: An organization system whose schemes and structures work together consistently, not as a random mix.

- Avoid **mixing schemes** inside one category list (hybrid schemes confuse users, e.g. topic + audience + task in one menu).
- Use **different schemes in separate, clearly marked areas** instead of blending them.
- Base design on three factors: **users, content, context** (business goals).
- Keep **consistency** in grouping, labels and structure across the whole site.
- Validate with **card sorting** and task-based user testing.
- **Cohesion** gives users a clear mental model: once they learn how one area works, they can predict the others.
- Organization system, navigation system and labeling system must **support each other**.
- Check cohesion with a **content inventory** and by walking through typical user tasks.
- **Governance:** assign ownership and rules so new content fits the system as the site grows.

3.5 Organizing the WWW

Definition: Applying organization schemes to the Web, which has no single central organization system.

- Web is huge, uncontrolled and heterogeneous: no standard vocabulary, duplicates, variable quality.
- **Directories:** human-edited hierarchical category lists (e.g. Yahoo Directory).
- **Search engines:** crawlers index pages automatically; users search by keyword.
- **Portals, bookmarks, social tagging (folksonomy)** also help organize and find content.
- **Hyperlinks** are the basic organizing/navigation mechanism.
- **Directories vs search engines:** directories give browsing through human-chosen categories (good quality, small coverage); search engines give very large coverage by automatic indexing (variable relevance).
- **Metadata** (descriptive tags) helps machines and users classify pages.

- **Problems:** no governance, spam, broken links, duplicate and outdated pages.
- **Social tagging (folksonomy):** users add their own tags; flexible and current, but uncontrolled and inconsistent.
- Lesson for designers: do not depend on users arriving at the home page; every page must stand on its own.

3.6 Browser Navigation Features

Definition: Built-in browser tools that help users move through and keep track of websites.

- **Back / Forward:** retrace the path; **Home:** return to start page; **Reload / Stop.**
- **Address bar (URL):** shows location; **History** and **Bookmarks/Favorites:** revisit pages.
- **Link colours** (visited / unvisited) and **status bar** (shows link URL); page **title / tabs.**
- **Design implications:** do not break the Back button (avoid forced redirects and unwanted pop-ups); give every page a meaningful title and a unique URL; keep visited-link colour.
- **Back button** is among the most used controls; sites that open new windows, use forced redirects or break history frustrate users.
- **Bookmarking** needs stable, meaningful URLs; dynamic or frame-based pages may bookmark the wrong page.
- Browser features are **outside the designer's control**, so design the site to work with them, not against them.
- Tabs and page titles make multiple open pages easy to identify.

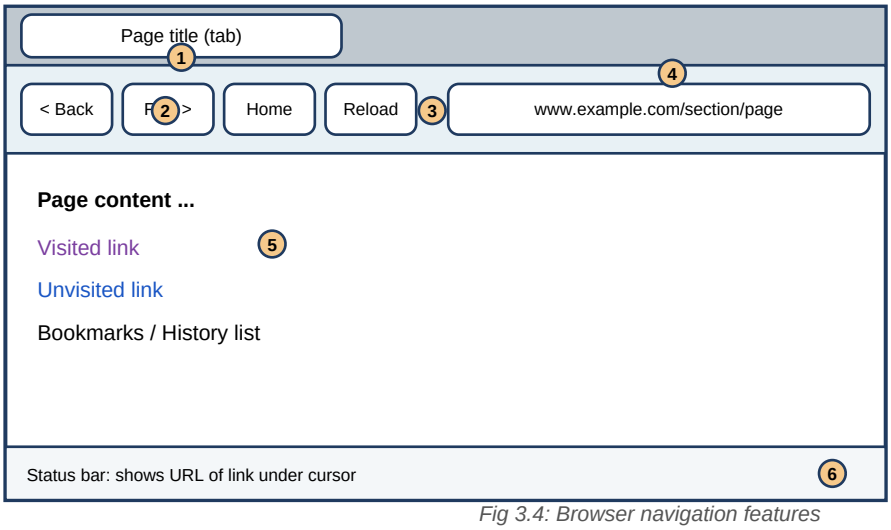


Fig 3.4: Browser navigation features

3.7 Building Context

Definition: Giving users cues about where they are, where they have been and where they can go, so they do not feel lost.

- **Why:** hyperlinked sites let users enter at any page, so every page must orient them.
- **Techniques:** site name/logo on every page, clear page titles, **breadcrumbs**, highlighting the current section ("you are here"), consistent layout, visited-link colours.
- Context reduces disorientation ("lost in hyperspace") and mental effort.
- **Sense of place:** consistent logo, colours and layout tell users they are still in the same site.
- **Colour coding or labels per section** show which part of the site users are in.

- **Breadcrumb types:** location-based (shows hierarchy position), path-based (pages visited), attribute-based (shows item properties).
- **Intranets:** show the organization or department name so employees know which unit owns the page.
- Context also means **explaining the page's purpose** with a clear title and heading.

Fig 3.5: Three questions context must answer

3.8 Improving Flexibility

Definition: Providing multiple routes and shortcuts so different users can reach content in their own way.

- A pure hierarchy (only up/down) is rigid: add **lateral links**, contextual links, shortcuts.
- Offer **several access paths**: menus, search, index, sitemap, related links.
- **Trade-off:** more flexibility means more links and possible clutter, so balance flexibility against simplicity and consistency.
- **Techniques:** shortcuts and quick links, fat or mega menus, search box on every page, site index and sitemap, multiple entry points.
- **Avoid dead ends:** every page should link onward (home, up, related).
- **Different users:** novices prefer simple menus; experts prefer shortcuts and search.
- Flexibility works best when it is **layered**: a simple main path, with optional extra routes.

3.9 Types of Navigation Systems

Definition: Sets of elements that let users move through content; they complement the organization system.

- **Integrated (embedded) navigation:** built into the page itself (Global, Local, Contextual).
- **Remote (supplemental) navigation:** separate tools outside the main hierarchy (Sitemap, Index, Guide, Search).
- Navigation is about **movement**; organization is about **grouping**; both work with labeling.
- Users find information by **browsing** (navigation), **searching** (search engine) or **asking** (guides, help).
- **Remote tools do not replace** integrated navigation; they add extra access routes.
- Navigation must answer: Where am I? Where can I go? How do I get back?

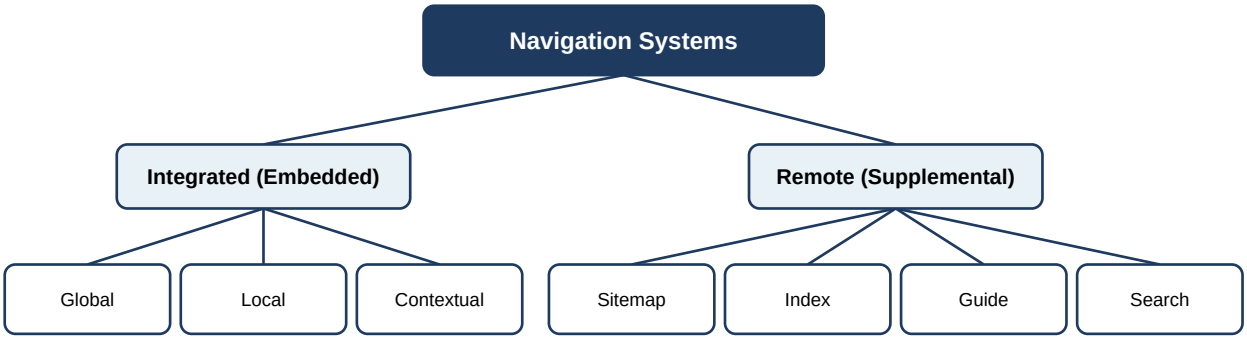


Fig 3.6: Types of navigation systems

3.10 Integrated Navigation Elements

Definition: Navigation elements embedded within the web pages themselves.

- **Global navigation guidelines:** always link to Home; include Search and Contact; keep the same position on every page; limit the number of items.
- **Local navigation:** shows siblings, parent and child pages of the current section; helps users explore deeper.
- **Contextual navigation:** editorial, hand-picked links; high value but costly to maintain; use descriptive link text, not “click here”.
- **Breadcrumbs** show the path from home to the current page and let users jump up levels.
- **Other integrated aids:** footer links, tabs, drop-down menus, “back to top” links.

Element	Description	Example
Global	Same on every page; gives access to main sections; ensures consistency.	Top menu bar: Home, Products, Contact, Search
Local	Complements global; moves users within a sub-section/sub-site.	Side menu of a department page
Contextual	Links placed inside content for related, ad hoc connections; flexible but can distract.	In-text links, “See also”, related articles

- Breadcrumbs and consistent placement of the global bar strengthen orientation.

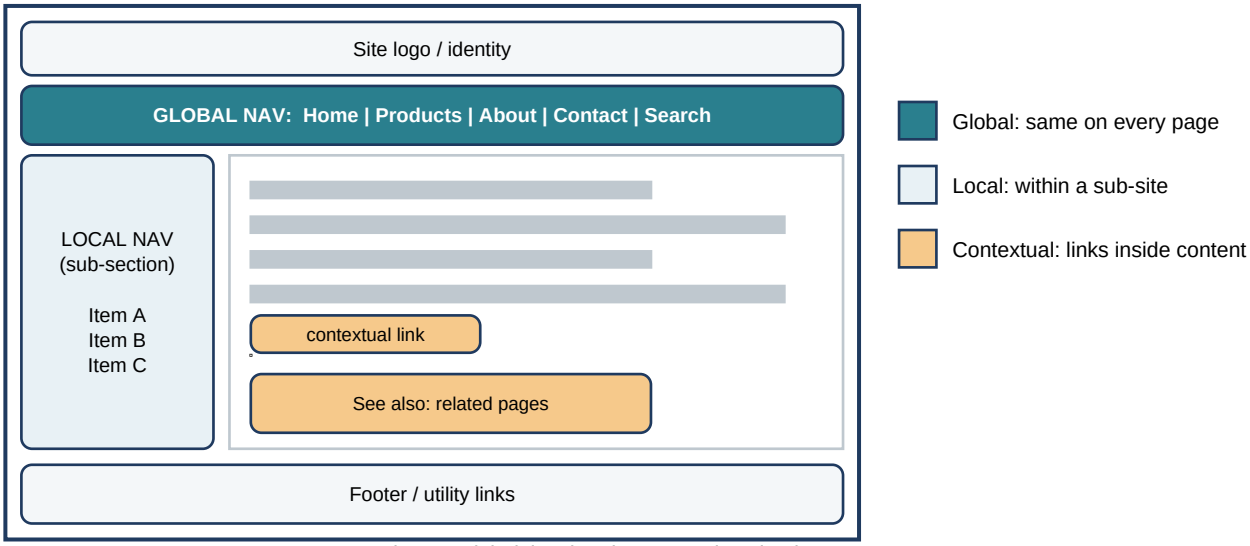


Fig 3.7: Global, local and contextual navigation on a page

3.11 Remote Navigation Elements

Definition: Supplemental navigation tools that sit outside the content hierarchy and give another view or access path.

- **Sitemap tips:** keep it to one or two levels; link to it from every page; keep it up to date; show the real structure.
- **Index tips:** arrange alphabetically; use terms users search for; add “see also” links; works best for small and medium sites.
- **Guide types:** guided tour (overview), tutorial (teaches steps), FAQ or help; useful for first-time users.
- **Search:** put the search box in a consistent place; show clear result lists; handle spelling and no-result cases.
- Remote navigation is **supplemental**: it helps users who cannot find content through the main structure.

Element	Purpose
Sitemap	Overview (text or graphic) of the site structure; shows content at a glance.
Index	Alphabetical list of keywords/terms linked to pages; useful for known-item access.
Guide	Tour, tutorial or step-by-step path introducing the site or a topic.
Search	Search box/engine to query content directly; used with large sites.

Fig 3.8: Remote navigation tools

3.12 Designing Elegant Navigation Systems

Definition: A navigation system that is simple, consistent and efficient, balancing flexibility against clutter.

- **Combine** global, local and contextual navigation; add remote tools (sitemap, index, search) for large sites.
- **Consistency:** same position, labels and behaviour on every page.
- **Simplicity:** do not overload pages with links; use clear, short labels.
- **Context:** always show current location; support the browser Back button.
- **User-centred:** design for audience tasks and behaviour (searching vs browsing); test and refine.
- **Evaluate navigation** with user tasks: can users find key items in few clicks and without getting lost?
- **Tune the amount of navigation** to the page: the home page needs broad choices; deeper pages need local and contextual links.
- **Accessibility:** text links, clear focus, usable on small screens.
- **Principle:** clear, small set of well-labeled choices is better than many unclear options.

UNIT IV: Labeling Systems

4.1 Labeling Systems

Definition: A **label** is a word, phrase or icon that represents a chunk of information; a **labeling system** is a consistent set of labels used across a site.

- Labels **represent** content, saving users from reading everything.
- Appear everywhere: headings, links, menus, buttons, index terms.
- Good labels speak the **language of the audience**, reduce thinking effort and build trust.
- **Why labels are hard:** language is ambiguous; one concept can have many words and one word many meanings.
- Labels act as the **signs of an information architecture**: they tell users what lies behind each link or heading.
- Poor labels cause users to **miss content** even when it exists.
- Labeling connects closely with organization and navigation: headings, menu items and search terms all come from the same label set.

4.2 Labeling Systems, Not Labels

Definition: Labels must be designed together as a system, not one at a time, since each label affects how others are understood.

- Aim for a **consistent** set across the site. Main consistency issues:
- **Why systems:** individual labels may look fine alone but conflict when seen together on one page.
- **Example:** a menu mixing “Search”, “Find a Store” and “Locations” has inconsistent syntax and overlapping meaning.
- **Rule:** decide style, syntax and vocabulary rules first, then write the labels.

Issue	Meaning / Example
Style	Punctuation, capitalization (e.g. "Contact Us" vs "contact us")
Presentation	Font, size, colour, spacing of labels
Syntax	Same grammatical form: all verbs (View, Edit) or all nouns
Granularity	Same level of detail across labels at one level
Comprehensiveness	Covers the full scope of content, no gaps
Audience	Same vocabulary level; avoid mixing jargon and plain words

4.3 Types of Labeling Systems

Definition: Categories of labels classified by where and how they appear.

- **Contextual links:** avoid vague text such as “click here”; use descriptive phrases that tell the destination.
- **Headings:** should be short, parallel in wording, and reflect hierarchy (H1, H2, H3).
- **Navigation labels:** prefer familiar words and few words; consistent with page titles.
- **Index terms:** manual indexing gives quality; synonyms and variants are mapped to a preferred term.

Fig 4.1: Types of labeling systems

Type	Key points
Contextual links	Label = link text inside content; must match what the target page is about; meaning depends on surrounding text.
Headings	Describe the content that follows; hierarchy of heading levels (size/colour) shows structure.
Navigation system choices	Labels in menus/bars (Home, About, Contact, Help, Search); short, consistent, standard where possible.
Index terms	Keywords/descriptive metadata for searching and browsing; use controlled vocabularies and thesauri for consistency.

4.4 Creating Effective Labeling Systems

Definition: Designing labels that are clear, consistent and understood by the target audience.

- **Narrow the scope** wherever possible (label only the content that needs it).
- **Develop one label per concept**; avoid synonyms for the same thing.
- **Be consistent** (style, syntax, granularity) and **consider the audience.**
- **Sources of labels:** existing site content, comparable sites, controlled vocabularies/thesauri, search logs, user input (card sorting).
- **Scope and clarity:** a label for a large collection must be broader than one for a single page; avoid overlap.
- **Audience language:** use the words users use, not internal jargon or brand names.
- **Controlled vocabulary:** a fixed list of approved terms with synonym rings and a thesaurus (broader, narrower, related terms).
- **Methods to gather labels:** content analysis, search-log analysis, competitor review, card sorting, user interviews.

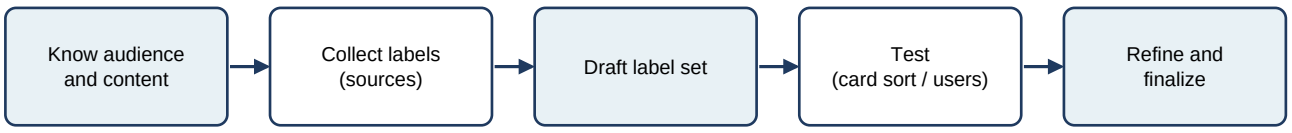


Fig 4.2: Process of creating a labeling system

4.5 Fine-Tuning the Labeling System

Definition: Reviewing and refining the draft label set until it is consistent and usable.

- Sort and **edit** the label list; remove duplicates; fix inconsistent style/syntax.
- Check that each label has **one meaning** and clearly differs from the others.

- Write a **style guide** and keep a controlled vocabulary for future content.
- **Test** with real users, analyse search logs, and **iterate**; labels must be maintained as content changes.
- **Review list:** check label length, ambiguity, abbreviations, and terms with several meanings.
- **Test methods:** card sorting (do users group as expected), closed card sort (fit items into your labels), usability tests (can users find items with these labels).
- **Log analysis:** look at failed searches to find terms users expect but you do not use.
- Fine-tuning is **ongoing**: new content and new terminology require regular updates.

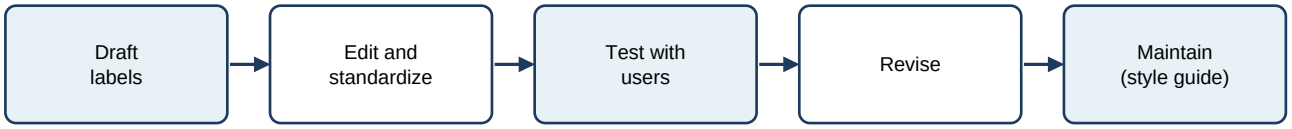


Fig 4.3: Fine-tuning cycle

4.6 Non-Representational Labeling Systems (Iconic Labels)

Definition: Labels that use icons or symbols instead of words, and do not state their meaning in text.

- **Advantages:** save space, visually attractive, quick for universal symbols (home, search, print, cart), less language-dependent.
- **Disadvantages:** ambiguous, culture-dependent, meaning must be learned; hard to represent abstract concepts.
- **Guidelines:** use standard/familiar icons, **pair icons with text**, keep them simple and consistent, test with users.
- **Used for:** toolbars, buttons, mobile apps, status indicators and international audiences.
- **Risk:** unfamiliar icons force users to guess or hover for tooltips.
- **Best practice:** universal icons can stand alone; all others should have a text label.
- Icons need **consistent style** (size, line weight, colour) to form a coherent system.

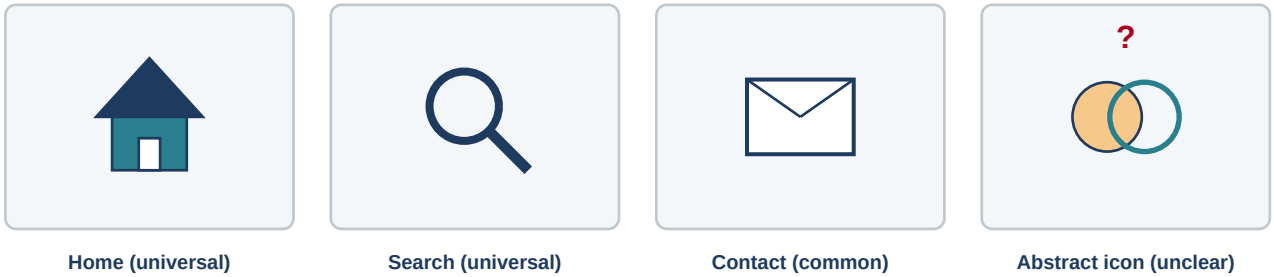


Fig 4.4: Icon labels: universal vs ambiguous

4.7 Double Challenge

Definition: The two-fold difficulty of icon labels: the icon must be well designed **and** correctly interpreted.

- **Challenge 1 (design):** create a simple, recognizable image.
- **Challenge 2 (meaning):** make every user, across cultures and experience levels, understand what it represents.
- **Hence:** text labels are usually safer than icons; icons work best when universal, or combined with text.

- **Why harder than text:** words carry shared dictionary meanings; icons depend on culture and prior experience.
- Designing a recognizable image **and** making it communicate the right idea is a double burden on the designer.
- Always **test icons** with the real audience before use.

Important Exam Questions

Unit III

- Define organization system. Differentiate organization schemes and organization structures.
- Explain the four organizational challenges (ambiguity, heterogeneity, perspective, politics).
- Compare exact and ambiguous organization schemes with examples.
- Explain hierarchy, database and hypertext structures with diagrams.
- What is a cohesive organization system? Why avoid hybrid schemes?
- How is the WWW organized? Compare directories and search engines.
- List and explain browser navigation features and their design implications.
- What is context? Explain techniques to build context.
- How can flexibility be improved in a navigation system?
- Explain integrated and remote navigation. Draw a diagram of global, local and contextual navigation.
- Describe sitemap, index, guide and search as remote navigation elements.
- How do you design an elegant navigation system?

Unit IV

- Define label and labeling system. Why are labeling systems important?
- Explain the consistency issues: style, presentation, syntax, granularity, comprehensiveness, audience.
- Explain the types of labeling systems with examples.
- How do you create an effective labeling system? Give sources of labels.
- Explain fine-tuning of a labeling system.
- What are non-representational (iconic) labels? List advantages and disadvantages.
- Explain the double challenge of icon labels.

Quick Definitions

Term	One-line definition
Organization scheme	Shared characteristic used to group content
Organization structure	Relationships between content items and groups
Hierarchy	Top-down tree of parent and child categories
Granularity	Level of detail of an information unit
Breadcrumb	Trail showing the path from home to current page
Global navigation	Menu present on every page
Local navigation	Menu for one section or sub-site
Contextual navigation	Links embedded within content
Sitemap	Overview of the site structure
Controlled vocabulary	Approved list of terms for consistent labeling
Label	Word or icon that represents content